F18CF – Linear Algebra

Tutorial Sheet 9: Diagonalisability, Dynamical Systems

Getting started.

1. (i) The characteristic polynomial is

$$\chi_A(\lambda) = \det \begin{pmatrix} 2-\lambda & 1 \\ 4 & -1-\lambda \end{pmatrix} = (2-\lambda)(-1-\lambda) - 4 = \lambda^2 - \lambda - 6 = (\lambda-3)(\lambda+2),$$

whose roots are $\lambda = 3, -2$. This 2×2 matrix has two distinct eigenvalues, hence it is diagonalisable.

• A vector $(x, y)^T$ belongs to the eigenspace $E_3(A)$ if and only if

$$\begin{cases} 2x + y &= 3x \\ 4x - y &= 3y \end{cases} \quad \text{i.e.} \quad \begin{cases} -x + y &= 0 \\ x - y &= 0 \end{cases}$$

i.e. if and only if $x = y$. Hence a basis for the eigenspace $E_3(A)$ is given by the vector $(1, 1)^T$.

• A vector $(x, y)^T$ belongs to the eigenspace $E_{-2}(A)$ if and only if

$$\begin{cases} 2x + y &= -2x \\ 4x - y &= -2y \end{cases} \quad \text{i.e.} \quad \begin{cases} 4x + y &= 0 \\ 4x + y &= 0 \end{cases}$$

i.e. $y = -4x$. Hence a basis for the eigenspace $E_{-2}(A)$ is given by the vector $(1, -4)^T$.
$\{(1, 1)^T, (1, -4)^T\}$ is a basis of $\mathbb{R}^2$ consisting of eigenvectors of A.

(ii) The characteristic polynomial of A is

$$\begin{aligned} \chi_A(\lambda) = \det \begin{pmatrix} 2-\lambda & -2 & 3 \\ 1 & 1-\lambda & 1 \\ 1 & 3 & -1-\lambda \end{pmatrix} &= (2-\lambda)[(1-\lambda)(-1-\lambda) - 3] + 2(-2-\lambda) + 3(2+\lambda) \\ &= \lambda^3 - 2\lambda^2 - 5\lambda + 6. \end{aligned}$$

We see that $\lambda = 1$ is a root of χ_A, so we can factorize χ_A:

$$\chi_A(\lambda) = (\lambda-1)(\lambda^2 - \lambda - 6) = (\lambda-1)(\lambda-3)(\lambda+2).$$

Hence the eigenvalues of A are $\lambda = -2$, $\lambda = 1$ and $\lambda = 3$. Since A is a 3×3 matrix with three distinct eigenvalues, it is diagonalisable.

• A vector $(x, y, z)^T$ belongs to the eigenspace $E_{-2}(A)$ if and only if

$$\begin{cases} 2x - 2y + 3z &= -2x \\ x + y + z &= -2y \\ x + 3y - z &= -2z \end{cases} \quad \text{i.e.} \quad \begin{cases} 4x - 2y + 3z &= 0 \\ x + 3y + z &= 0 \\ x + 3y + z &= 0 \end{cases}$$

We compute a basis of the eigenspace via Gaussian elimination:

$$\left(\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 4 & -2 & 3 & 0 \end{array}\right) \overset{R_2 \to R_2 - 4R_1}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 3 & 1 & 0 \\ 0 & -14 & -1 & 0 \end{array}\right)$$

There is one free variable $z = \alpha$, and we get by substitution $y = -\frac{1}{14}\alpha$ and $x = -\frac{11}{14}\alpha$. Thus $(x, y, z)^T = \alpha(-11/14, -1/14, 1)^T$, and a basis of the eigenspace $E_{-2}(A)$ is given by $(-11, -1, 14)^T$.

• A vector $(x, y, z)^T$ belongs to the eigenspace $E_1(A)$ if and only if

$$\left\{\begin{array}{rcl} 2x - 2y + 3z &=& x \\ x + y + z &=& y \\ x + 3y - z &=& z \end{array}\right. \quad \text{i.e.} \quad \left\{\begin{array}{rcl} x - 2y + 3z &=& 0 \\ x + z &=& 0 \\ x + 3y - 2z &=& 0 \end{array}\right.$$

We compute a basis of the eigenspace via Gaussian elimination:

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 3 & -2 & 0 \end{array}\right) \overset{\substack{R_2 \to R_2 - R_1 \\ R_3 \to R_3 - R_1}}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 5 & -5 & 0 \end{array}\right) \overset{R_3 \to 2R_3 - 5R_2}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

There is one free variable $z = \alpha$, and we get by substitution $y = \alpha$ and $x = -\alpha$ and so $(x, y, z)^T = \alpha(-1, 1, 1)^T$.
Hence a basis of $E_1(A)$ is given by the vector $(-1, 1, 1)^T$.
• A vector $(x, y, z)^T$ belongs to the eigenspace $E_3(A)$ if and only if

$$\left\{\begin{array}{rcl} 2x - 2y + 3z &=& 3x \\ x + y + z &=& 3y \\ x + 3y - z &=& 3z \end{array}\right. \quad \text{i.e.} \quad \left\{\begin{array}{rcl} -x - 2y + 3z &=& 0 \\ x - 2y + z &=& 0 \\ x + 3y - 4z &=& 0 \end{array}\right.$$

We compute a basis of the eigenspace via Gaussian elimination:

$$\left(\begin{array}{ccc|c} -1 & -2 & 3 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & 3 & -4 & 0 \end{array}\right) \overset{\substack{R_2 \to R_2 + R_1 \\ R_3 \to R_3 + R_1}}{\rightsquigarrow} \left(\begin{array}{ccc|c} -1 & -2 & 3 & 0 \\ 0 & -4 & 4 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right) \overset{R_3 \to 4R_3 + R_2}{\rightsquigarrow} \left(\begin{array}{ccc|c} -1 & -2 & 3 & 0 \\ 0 & -4 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

There is one free variable $z = \alpha$, and we get by substitution $y = \alpha$ and $x = \alpha$ and so $(x, y, z)^T = \alpha(1, 1, 1)^T$.
Hence a basis of the eigenspace $E_3(A)$ is given by the vector $(1, 1, 1)^T$.
$\{(-11, -1, 14)^T, (-1, 1, 1)^T, (1, 1, 1)^T\}$ is a basis for $\mathbb{R}^3$ consisting of eigenvectors of A.

(iii) The characteristic polynomial of A is

$$\begin{aligned} \chi_A(\lambda) &= \det \begin{pmatrix} 2 - \lambda & -2 & 3 \\ 10 & -4 - \lambda & 5 \\ 5 & -4 & 6 - \lambda \end{pmatrix} \\ &= (2 - \lambda)[(-4 - \lambda)(6 - \lambda) + 20] + 2[10(6 - \lambda) - 25] + 3[-40 + 5(4 + \lambda)] \\ &= \lambda^3 - 4\lambda^2 + 5\lambda - 2. \end{aligned}$$

We see that $\lambda = 1$ is a root of χ_A, so we factorize χ_A further:

$$\chi_A(\lambda) = (\lambda - 1)(\lambda^2 - 3\lambda + 2) = (\lambda - 1)^2(\lambda - 2).$$

Hence the eigenvalues are $\lambda = 1$ and $\lambda = 2$.
• A vector $(x, y, z)^T$ belongs to the eigenspace $E_1(A)$ if and only if

$$\begin{cases} 2x - 2y + 3z &= x \\ 10x - 4y + 5z &= y \\ 5x - 4y + 6z &= z \end{cases} \quad \text{i.e.} \quad \begin{cases} x - 2y + 3z &= 0 \\ 10x - 5y + 5z &= 0 \\ 5x - 4y + 5z &= 0 \end{cases}$$

We compute a basis of the eigenspace via Gaussian elimination:

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 2 & -1 & 1 & 0 \\ 5 & -4 & 5 & 0 \end{array}\right) \overset{\substack{R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 5R_1}}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 0 & 3 & -5 & 0 \\ 0 & 6 & -10 & 0 \end{array}\right) \overset{R_3 \to R_3 - 2R_2}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & -2 & 3 & 0 \\ 0 & 3 & -5 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

There is one free variable $z = \alpha$, and we get by substitution $y = \frac{5}{3}\alpha$ and $x = \frac{1}{3}\alpha$.
Hence the eigenvectors corresponding to $\lambda = 1$ are of the form $\alpha(\frac{1}{3}, \frac{5}{3}, 1)^T$, and it follows that a basis of the eigenspace $E_1(A)$ is given by the vector $(1, 5, 3)^T$.
• A vector $(x, y, z)^T$ belongs to the eigenspace $E_2(A)$ if and only if

$$\begin{cases} 2x - 2y + 3z &= 2x \\ 10x - 4y + 5z &= 2y \\ 5x - 4y + 6z &= 2z \end{cases} \quad \text{i.e.} \quad \begin{cases} -2y + 3z &= 0 \\ 10x - 6y + 5z &= 0 \\ 5x - 4y + 4z &= 0 \end{cases}$$

We find a basis of the eigenspace via Gaussian elimination:

$$\left(\begin{array}{ccc|c} 5 & -4 & 4 & 0 \\ 10 & -6 & 5 & 0 \\ 0 & -2 & 3 & 0 \end{array}\right) \overset{R_2 \to R_2 - 2R_1}{\rightsquigarrow} \left(\begin{array}{ccc|c} 5 & -4 & 4 & 0 \\ 0 & 2 & -3 & 0 \\ 0 & -2 & 3 & 0 \end{array}\right) \overset{R_2 \to R_2 + R_1}{\rightsquigarrow} \left(\begin{array}{ccc|c} 5 & -4 & 4 & 0 \\ 0 & 2 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

There is one free variable $z = \alpha$, and we get by substitution $y = \frac{3}{2}\alpha$ and $x = \frac{2}{5}\alpha$.
Hence the eigenvectors corresponding to $\lambda = 2$ are of the form $\alpha(\frac{2}{5}, \frac{3}{2}, 1)^T$, and it follows that a basis of the eigenspace $E_2(A)$ is given by the vector $(4, 15, 10)^T$.
Since A is a 3×3 matrix with eigenvalues $1, 2$ but dim $E_1(A)$+ dim $E_2(A) < 3$, it follows that A is not diagonalisable.

2. (a) The characteristic polynomial of A is

$$\chi_A(\lambda) = \det \begin{pmatrix} 1 - \lambda & 3 \\ 1 & -1 - \lambda \end{pmatrix} = (1 - \lambda)(-1 - \lambda) - 3 = \lambda^2 - 4.$$

So the eigenvalues of A are ± 2. Since A is a 2×2 matrix with two distinct eigenvalues, it is diagonalisable.

• A vector $(x,y)^T$ belongs to the eigenspace $E_2(A)$ if and only if

$$\left\{ \begin{array}{rcl} x+3y & = & 2x \\ x-y & = & 2y \end{array} \right. \quad \text{i.e.} \quad \left\{ \begin{array}{rcl} x-3y & = & 0 \end{array} \right.$$

Thus, a basis of the corresponding eigenspace $E_2(A)$ is the vector $(3,1)^T$.

• A vector $(x,y)^T$ belongs to the eigenspace $E_{-2}(A)$ if and only if

$$\left\{ \begin{array}{rcl} x+3y & = & -2x \\ x-y & = & -2y \end{array} \right. \quad \text{i.e.} \quad \left\{ \begin{array}{rcl} x+y & = & 0 \end{array} \right.$$

Thus, a basis of the corresponding eigenspace $E_{-2}(A)$ is the vector $(1,-1)^T$.
We thus construct the matrix of change of basis

$$P = \begin{pmatrix} 1 & 3 \\ -1 & 1 \end{pmatrix},$$

whose inverse we find via Gaussian elimination (see Tutorial 5):

$$P^{-1} = \frac{1}{4} \begin{pmatrix} 1 & -3 \\ 1 & 1 \end{pmatrix}.$$

We thus have

$$A = P \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix} P^{-1}.$$

(b) We thus have

$$\begin{aligned} A^{10} &= (\ P \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix} P^{-1}\ )^{10} = P \begin{pmatrix} (-2)^{10} & 0 \\ 0 & 2^{10} \end{pmatrix} P^{-1} \\ &= P \times 2^{10} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} P^{-1} = \begin{pmatrix} 2^{10} & 0 \\ 0 & 2^{10} \end{pmatrix}. \end{aligned}$$

Problems.

3. **A prey-predator system.**

(a) We are studying the system

$$\begin{pmatrix} f_{n+1} \\ g_{n+1} \end{pmatrix} = \underbrace{\begin{pmatrix} 0.5 & 0.3 \\ -0.2 & 1.2 \end{pmatrix}}_{=A} \begin{pmatrix} f_n \\ g_n \end{pmatrix}.$$

The population after one year is given by

$$\begin{pmatrix} f_1 \\ g_1 \end{pmatrix} = A \begin{pmatrix} f_0 \\ g_0 \end{pmatrix} = \begin{pmatrix} 0.5 & 0.3 \\ -0.2 & 1.2 \end{pmatrix} \begin{pmatrix} 100 \\ 5000 \end{pmatrix} = \begin{pmatrix} 1550 \\ 5980 \end{pmatrix}.$$

The population after two years is given by

$$\begin{pmatrix} f_2 \\ g_2 \end{pmatrix} = A^2 \begin{pmatrix} f_0 \\ g_0 \end{pmatrix} = \begin{pmatrix} 0.19 & 0.51 \\ -0.34 & 1.38 \end{pmatrix} \begin{pmatrix} 100 \\ 5000 \end{pmatrix} = \begin{pmatrix} 2569 \\ 6866 \end{pmatrix}.$$

(b) The matrix associated to this system is

$$A = \begin{pmatrix} 0.5 & 0.3 \\ -0.2 & 1.2 \end{pmatrix}$$

and its characteristic polynomial is

$$\chi_A(\lambda) = (0.5 - \lambda)(1.2 - \lambda) + 0.06 = \lambda^2 - 1.7\lambda + 0.66,$$

whose eigenvalues are $\lambda_1 = 1.1$ and $\lambda_2 = 0.6$ (check that is coherent with the trace and the determinant!).

Since A is a 2×2 matrix with two distinct eigenvalues, it is diagonalisable.

(c) Since A is diagonalisable by (b), we choose a basis of eigenvectors $\vec{e}_1, \vec{e}_2$ for the eigenvalues λ_1, λ_2 respectively. We can write the initial condition vector $(f_0, g_0)^T$ as a linear combination of $\vec{e}_1, \vec{e}_2$:

$$\begin{pmatrix} f_0 \\ g_0 \end{pmatrix} = \alpha_1 \vec{e}_1 + \alpha_2 \vec{e}_2$$

for some constants $\alpha_1, \alpha_2 \in \mathbb{R}$. We thus have

$$\begin{pmatrix} f_n \\ g_n \end{pmatrix} = A^n \begin{pmatrix} f_0 \\ g_0 \end{pmatrix} = \alpha_1 A^n \vec{e}_1 + \alpha_2 A^n \vec{e}_2 = \alpha_1 \lambda_1^n \vec{e}_1 + \alpha_2 \lambda_2^n \vec{e}_2.$$

In particular, by taking each component of this vector, we see that there exist constants $a_1, a_2, b_1, b_2 \in \mathbb{R}$ (that depend only on $\alpha_1, \alpha_2, \vec{e}_1, \vec{e}_2$) such that

$$\begin{cases} f_n &= a_1 \lambda_1^n + a_2 \lambda_2^n \\ g_n &= b_1 \lambda_1^n + b_2 \lambda_2^n \end{cases}$$

We find the constants a_1, a_2 by evaluating at $n = 0$ and $n = 1$, which yields the system

$$\begin{cases} a_1 + a_2 &= 100 \\ 1.1a_1 + 0.6a_2 &= 1550 \end{cases},$$

which we solve as usual via Gaussian elimination, which yields $a_1 = 2980$, $a_2 = -2880$.

Similarly, we find the constants b_1, b_2 by evaluating at $n = 0$ and $n = 1$, which yields the system

$$\begin{cases} b_1 + b_2 &= 5000 \\ 1.1b_1 + 0.6b_2 &= 5980 \end{cases},$$

which we solve as usual via Gaussian elimination, which yields $b_1 = 5960$, $b_2 = -960$.

Thus, we have

$$\begin{cases} f_n &= 2980 \times (1.1)^n - 2880 \times (0.6)^n \\ g_n &= 5960 \times (1.1)^n - 960 \times (0.6)^n \end{cases},$$

(d) We have

$$\frac{g_n}{f_n} = \frac{5960 \times (1.1)^n - 960 \times (0.6)^n}{2980 \times (1.1)^n - 2880 \times (0.6)^n} = 2 \times \frac{1 - 960/5960(0.6/1.1)^n}{1 - 2880/5960(0.6/1.1)^n} \to_\infty 2.$$

Thus, the population of frogs and gnats explodes, and asymptotically the gnat population grows twice as fast as the frog population.

4. **A sequence satisfying a linear recurrence.**

(a) We have

$$\begin{pmatrix} u_{n+1} \\ u_{n+2} \\ u_{n+3} \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1/4 & 1/4 & 1 \end{pmatrix}}_{=A} \begin{pmatrix} u_n \\ u_{n+1} \\ u_{n+2} \end{pmatrix}.$$

(b) We have

$$\chi_A(\lambda) = \det \begin{pmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ -1/4 & 1/4 & 1-\lambda \end{pmatrix} = -\lambda^3 + \lambda^2 + \frac{1}{4}\lambda - \frac{1}{4}.$$

We notice that $\lambda = 1$ is a root of χ_A, so we can factorise $(\lambda - 1)$. We find:

$$\chi_A(\lambda) = (\lambda - 1)(-\lambda^2 + \frac{1}{4}) = (\lambda - 1)(\frac{1}{2} - \lambda)(\frac{1}{2} + \lambda).$$

Thus, A is a 3×3 matrix with exactly three distinct eigenvalues: $1, 1/2, -1/2$, and it follows that A is diagonalisable.

(c) Since A is diagonalisable with eigenvalues $1, 1/2, -1/2$ by (b), there exists a basis of eigenvectors $\vec{e}_1, \vec{e}_2, \vec{e}_3$ for the eigenvalues $1, 1/2, -1/2$ respectively. We write the initial vector $\vec{x}_0$ as a linear combination of this basis:

$$\vec{x}_0 = \alpha_1\vec{e}_1 + \alpha_2\vec{e}_2 + \alpha_3\vec{e}_3$$

for some $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$, and we get

$$\vec{x}_n = A^n\vec{x}_0 = \alpha_1 A^n\vec{e}_1 + \alpha_2 A^n\vec{e}_2 + \alpha_3 A^n\vec{e}_3 = \alpha_1\vec{e}_1 + \alpha_2(\frac{1}{2})^n\vec{e}_2 + \alpha_3(-\frac{1}{2})^n\vec{e}_3.$$

By considering the first component of that vector, it follows that there exist constants $a_1, a_2, a_3 \in \mathbb{R}$ (that depend only on $\alpha_1, \alpha_2, \alpha_3$ and $\vec{e}_1, \vec{e}_2, \vec{e}_3$) such that

$$u_n = a_1 + a_2(1/2)^n + a_3(-1/2)^n$$

for all n. We find these constants by evaluating at $n = 0, 1, 2$, which yields the system

$$\begin{cases} a_1 + a_2 + a_3 &= 0 \\ a_1 + \frac{1}{2}a_2 - \frac{1}{2}a_3 &= 1 \\ a_1 + \frac{1}{4}a_2 + \frac{1}{4}a_3 &= 2 \end{cases},$$

which we solve as usual via Gaussian elimination, yielding $a_1 = 8/3, a_2 = -3, a_3 = 1/3$.

(d) We thus have

$$u_n = \frac{8}{3} - 3(\frac{1}{2})^n + \frac{1}{3}(-\frac{1}{2})^n \to_\infty \frac{8}{3}.$$

5. **Diagonalisability of real symmetric 2×2 matrices.**

(a) We have $\chi_A(\lambda) = (a-\lambda)(c-\lambda) - b^2 = \lambda^2 - (a+c)\lambda + (ac - b^2)$.
The discriminant of χ_A is $\Delta = (a+c)^2 - 4(ac-b^2) = a^2 - 2ac + c^2 + 4b^2 = (a-c)^2 + 4b^2$.
Since $b \neq 0$, we have $\Delta > 0$. Thus, χ_A has exactly two distinct roots, hence A is a 2×2 matrix with exactly two distinct eigenvalues, hence it is diagonalisable.

(b) In matrix form, we have

$$\langle \vec{x}, A\vec{y} \rangle = \vec{x}^T A \vec{y}$$

and, since $A^T = A$,

$$\langle A\vec{x}, \vec{y} \rangle = (A\vec{x})^T \vec{y} = \vec{x}^T A^T \vec{y} = \vec{x}^T A \vec{y},$$

hence the equality.

(c) We now apply the previous equality for $\vec{x} = \vec{x}_1$ and $\vec{y} = \vec{x}_2$, which yields $\langle A\vec{x}_1, \vec{x}_2 \rangle = \langle \vec{x}_1, A\vec{x}_2 \rangle$. But since $A\vec{x}_1 = \lambda_1 \vec{x}_1$ and $A\vec{x}_2 = \lambda_2 \vec{x}_2$, we get

$$\lambda_1 \langle \vec{x}_1, \vec{x}_2 \rangle = \lambda_2 \langle \vec{x}_1, \vec{x}_2 \rangle.$$

Since $\lambda_1 \neq \lambda_2$, this implies that $\langle \vec{x}_1, \vec{x}_2 \rangle = 0$.
Since $\vec{x}_1, \vec{x}_2$ are eigenvectors for distinct eigenvalues, they are linearly independent by a result from the course. Since we have a linearly independent family of 2 vectors in $\mathbb{R}^2$, it is a basis. Since these two vectors are orthogonal by the above, we thus have an orthogonal basis of $\mathbb{R}^2$.

(d) Let $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ be a real symmetric 2×2 matrix. If $b \neq 0$, then it follows from (c) that $\mathbb{R}^2$ admits an orthogonal basis made of eigenvectors of A . If $b = 0$, then A is diagonal, hence the standard basis of $\mathbb{R}^2$ is an orthogonal basis made of eigenvectors of A.

6. (a) We can reformulate the equation given as $f(\vec{x}) = f(\vec{0}) + \frac{1}{2}\langle \vec{x}, H\vec{x} \rangle +$ (higher terms). For a vector $\vec{x} \in \mathbb{R}^2$ of the form $\vec{x} = a_1\vec{v}_1 + a_2\vec{v}_2$ (with $a_1, a_2 \in \mathbb{R}$), we have

$$H\vec{x} = a_1 H\vec{v}_1 + a_2 H\vec{v}_2 = a_1\lambda_1\vec{v}_1 + a_2\lambda_2\vec{v}_2,$$

hence, using using the (bi-)linearity of the inner product and the fact that $\vec{v}_1, \vec{v}_2$ is an orthonormal basis of $\mathbb{R}^2$:

$$\begin{aligned} \vec{x}^T H\vec{x} = \langle \vec{x}, H\vec{x} \rangle &= \langle a_1\vec{v}_1 + a_2\vec{v}_2, a_1\lambda_1\vec{v}_1 + a_2\lambda_2\vec{v}_2 \rangle \\ &= \lambda_1 a_1^2 \langle \vec{v}_1, \vec{v}_1 \rangle + \lambda_2 a_1 a_2 \langle \vec{v}_1, \vec{v}_2 \rangle + \lambda_1 a_1 a_2 \langle \vec{v}_2, \vec{v}_1 \rangle + \lambda_2 a_2^2 \langle \vec{v}_2, \vec{v}_2 \rangle \\ &= \lambda_1 a_1^2 + \lambda_2 a_2^2. \end{aligned}$$

This gives the desired result.

(b) Since H is invertible, both λ_1 and λ_2 are non-zero. From the previous formula, we get three different cases to consider:

- If $\lambda_1, \lambda_2 > 0$, then $\langle \vec{x}, H\vec{x} \rangle \geq 0$ for all $\vec{x} \in \mathbb{R}^2$, hence $f(\vec{x}) \geq f(\vec{0})$ for $\vec{x}$ small enough. In particular, $\vec{0}$ is a local minimum.
- If $\lambda_1, \lambda_2 < 0$, then $\langle \vec{x}, H\vec{x} \rangle \leq 0$ for all $\vec{x} \in \mathbb{R}^2$, hence $f(\vec{x}) \leq f(\vec{0})$ for $\vec{x}$ small enough. In particular, $\vec{0}$ is a local maximum.
- If λ_1 and λ_2 have different signs, then $f(\alpha\vec{v}_1) > 0$ and $f(\alpha\vec{v}_2) < 0$ for small values of $\alpha \neq 0$. In particular, $\vec{0}$ is a saddle point.

(c) We have $\mathrm{tr}(H) = \lambda_1 + \lambda_2$ and $\det(H) = \lambda_1\lambda_2$. It follows from (b) that

$$\vec{0} \text{ is a saddle point } \Leftrightarrow \lambda_1, \lambda_2 \text{ have different signs } \Leftrightarrow \det(H) < 0$$

Conversely, $\vec{0}$ is a local extremum if and only if λ_1, λ_2 have the same sign, that is, if and only if $\det(H) > 0$. Moreover, the sign of λ_1, λ_2 is then given by the sign of $\lambda_1 + \lambda_2 = \mathrm{Tr}(H)$. We thus get:

$$\vec{0} \text{ is a local minimum } \Leftrightarrow \lambda_1, \lambda_2 > 0 \Leftrightarrow \det(H) > 0 \text{ and } \mathrm{Tr}(H) > 0$$

and

$$\vec{0} \text{ is a local maximum } \Leftrightarrow \lambda_1, \lambda_2 < 0 \Leftrightarrow \det(H) > 0 \text{ and } \mathrm{Tr}(H) < 0.$$